HEALTHCARE COVID-19 TRENDS ANALYSIS

Domain Executive Briefing: Healthcare Domain (healthcare_covid.csv) | Date: August 2026

1. Executive Summary

Analysis of regional COVID-19 infection trends, recovery vs. death rates, hospital bed utilization, and demographic impacts.

2. Key Performance Indicators (KPI Overview)

Metric Description	Key Performance Value	Strategic Benchmark / Context
Total New Cases	502,125	Cumulative across recorded period
Overall Recovery Rate	94.9%	Estimated recovery fraction
Most Impacted State	Delhi	Highest cumulative case load

3. Formal Statistical Hypothesis Testing

Vaccination vs Positivity: --- FORMAL HYPOTHESIS TEST --- H■: No correlation between vaccination rates and positivity. H■: Negative correlation exists. Test used: Pearson Correlation Test statistic (r): 0.0218 p-value: 0.49170 95% Confidence Interval: [-0.0403, 0.0836] Significance level: 0.05 Decision: Fail to Reject H■ Business interpretation: No statistically significant linear relationship was detected between vaccination doses and positivity rate. Why this matters: The result does not provide sufficient statistical evidence for a targeted intervention. ------------------------------

4. Core Analytical & Statistical Insights

Insight 1: Finding: 3 states carry 60% of the total caseload. Meaning: The pandemic impact is geographically concentrated.

Insight 2: Finding: No significant linear relationship was detected between vaccination doses and positivity. Meaning: Further multi-variate study needed.

5. Graphical Analytics & Visual Evidence

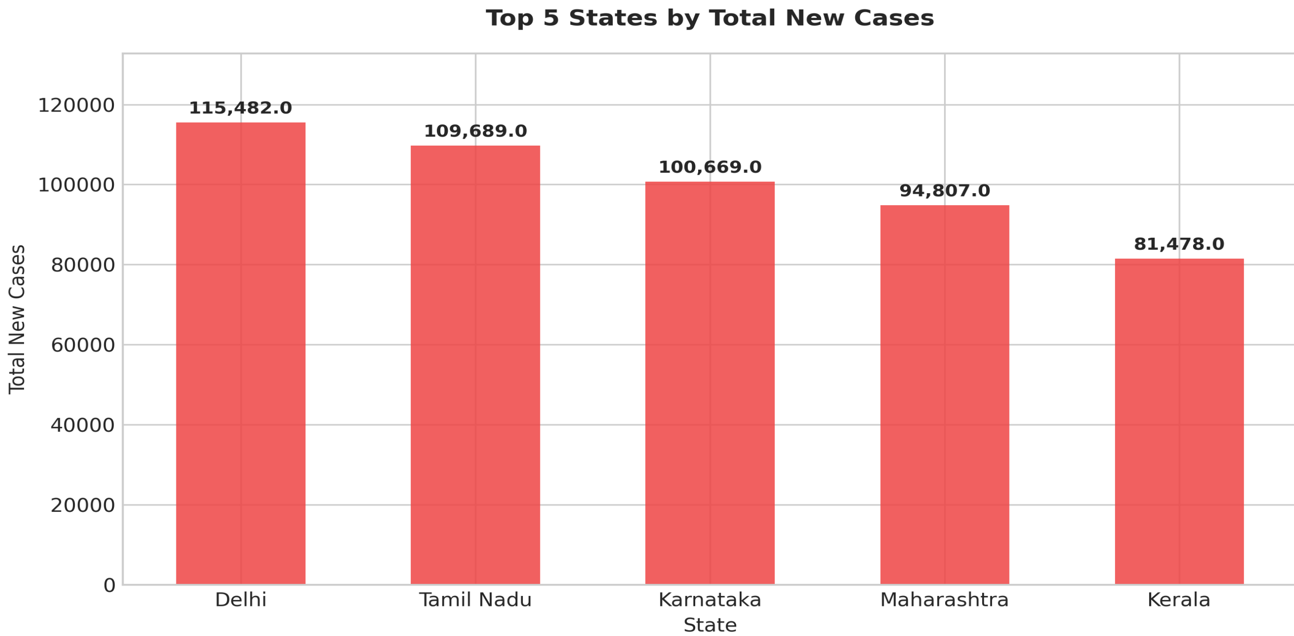

Figure 1: Visual analytical representation derived from Healthcare Domain (healthcare_covid.csv) dataset.

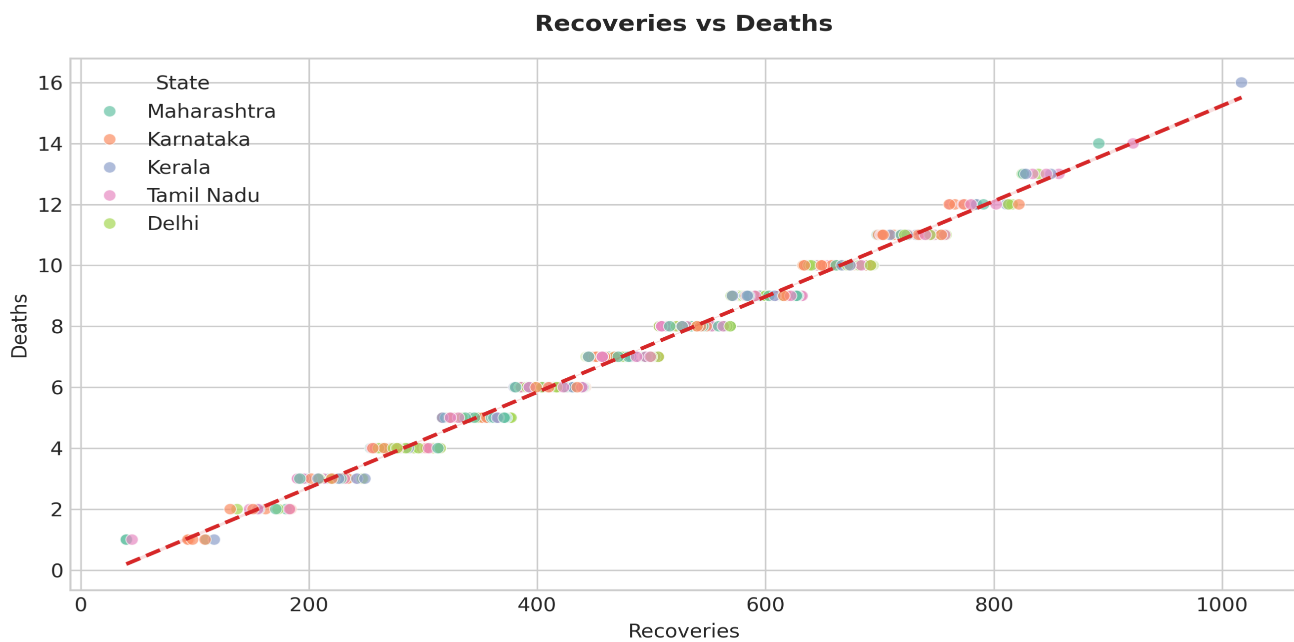

Figure 2: Visual analytical representation derived from Healthcare Domain (healthcare_covid.csv) dataset.

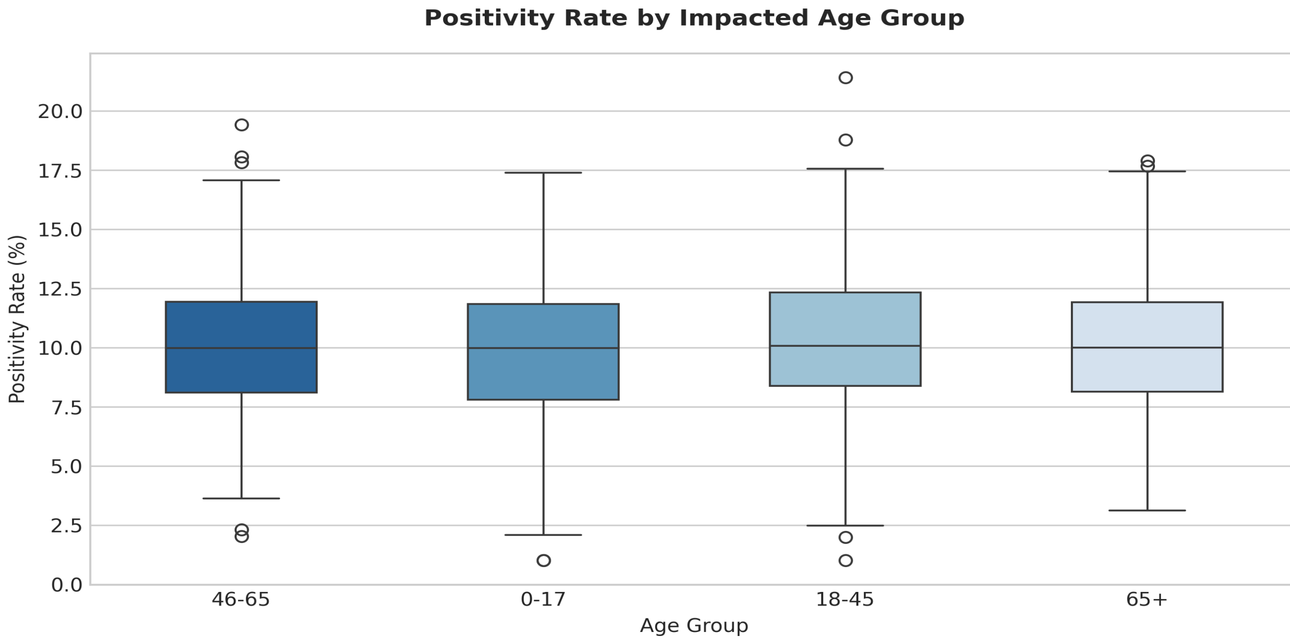

Figure 3: Visual analytical representation derived from Healthcare Domain (healthcare_covid.csv) dataset.

6. Strategic Recommendations & Action Plan

Action 1: Use state-level case concentration as one input for further healthcare resource-allocation analysis, alongside hospital capacity and population data.

Action 2: Consider targeted public health messaging campaigns specifically geared toward the 18-25 demographic.